

Brief summary of this function: Plots all of the subplots in the main live script at once for the different phases of Power vs Time.

Detailed explanation of this function: This live script helper function was made to make the readability of the main script easier, so there isn't numerous lines for plotting (ie: subplot, plot, title) in the main code. It is made to plot Power vs Time in the different phases. This function starts off by defining the modes in questions, than creating plots for them in one figure. It starts off with CC, CV, Rest, then Undefined phases.

```
function plotPower_phases(time, power, modes) %naming the function

%defining the different phases
CC = modes == "CC";
CV = modes == "CV";
Rest = modes == "Rest";
Undefined = modes == "Undefined";

%opening a figure for the plots
figure

%using basic plotting techniques used in other helper functions/main script
%ie: subplot, plot, title, labeling axes

% plotting the CC portion of the graph for power vs time (CC is the only
% thing on the first subplot)
subplot(2,2,1)
plot(time(CC), power(CC), 'Color', [0.5,0,0]) %choosing same dark red color
for consistency

title('Power vs Time (CC)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

%plotting the CV graph
subplot(2,2,2)
plot(time(CV), power(CV), 'b') %choosing same blue color as before for CV

title('Power vs Time (CV)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

%plotting the graph during the Rest phase
subplot(2,2,3)
plot(time(Rest), power(Rest), 'g') %choosing same green color for rest

title('Power vs Time (Rest)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on
```

```

%plotting for Undefined regions
subplot(2,2,4)
plot(time(Undefined), power(Undefined), 'c') %choosing same cyan color for
undefined

title('Power vs Time (Undefined)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

sgtitle('Power vs Time Graphs for the Seperate Phases') %main title for the
subplot

end

```